# TinyStar'Ex

## First-use guide for the spectrograph

Version 1.1

A STAROS project

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## 1 Purpose

This document briefly explains how to acquire your first spectrum with `TinyStar'Ex` (TSE).

If you have carried out the bench adjustments correctly, this should be a mere formality.

First and foremost, however, you need a well-adjusted telescope, even a modest one, and a mount that points accurately at celestial objects and that you know how to use. We assume that you have already observed with your telescope and, ideally, obtained sharp images of celestial objects with an electronic camera. These basics are taken as a prerequisite and are not covered in this document.

## 2 Mounting on the telescope

If possible, attach the spectrograph by its 50 mm nosepiece (figure 1).



**Figure 1.** TinyStar'Ex installed at the focus of a 250 mm f/4 Newtonian telescope.

Orient the spectrograph relative to the tube according to whether you wish to guide in right ascension or in declination. In the example of figure 2, guiding in declination was chosen. To guide in right ascension, rotate either the tube or the spectrograph in its nosepiece by 90° from the position shown in the figure.



**Figure 2.** Adjusting the orientation of the spectrograph.

Make sure that the axes of the spectrograph are accurately parallel or perpendicular to those of the mount, so that the movements of the mount make objects travel precisely along the rows and columns of the guiding sensor.

To work comfortably and efficiently (the two go hand in hand), you are strongly advised to equip yourself with three cameras: one acting as an electronic finder, one for pointing and guiding on TSE, and of course one for acquiring the spectra (figure 3). For the latter, a camera fitted with the Sony IMX585 sensor is currently an excellent choice: low noise, ideal pixel size, very low thermal signal, and an image format well suited to TSE. Any model can be used, however, including colour cameras, although their images are harder to process. If possible, choose a cooled-sensor model for this so-called ‘science’ camera.



**Figure 3.** General view of the telescope fitted with the TSE spectrograph and an electronic finder.

# 3 Acquisition software and pointing

You need efficient software to acquire technical, or even scientific, data. Steer clear of conventional imaging software: it may produce beautiful pictures, but very seldom good spectra.

Choose instead software that offers, in addition to the usual astrophotography functions, features designed for spectroscopy. You will find tools there that will change your life. Possible choices include *Prism* by Cyril Cavadore (commercial, but very reasonably priced given what it does and the usual investment in astronomical equipment) and *CCDciel* by Patrick Chevalley (free and efficient).

Here we use *CCDciel*, which is under very active development (see `https://buil.astrosurf.com/specinti4/guide_ccdciel.pdf`).

For pointing, a 'Goto' mount is all but essential and a great help in spectroscopy. If you also have a good, well-aligned electronic finder, the astrometry functions built into *Prism* and *CCDciel* will bring the target very close to the entrance slit, even if it is an obscure quasar that you can barely see, or not see at all!

In the example of figure 4, the target, here the star 7 Tri (= HD 13869), is selected in the *Carte du Ciel* software, the indispensable companion to *CCDciel*.

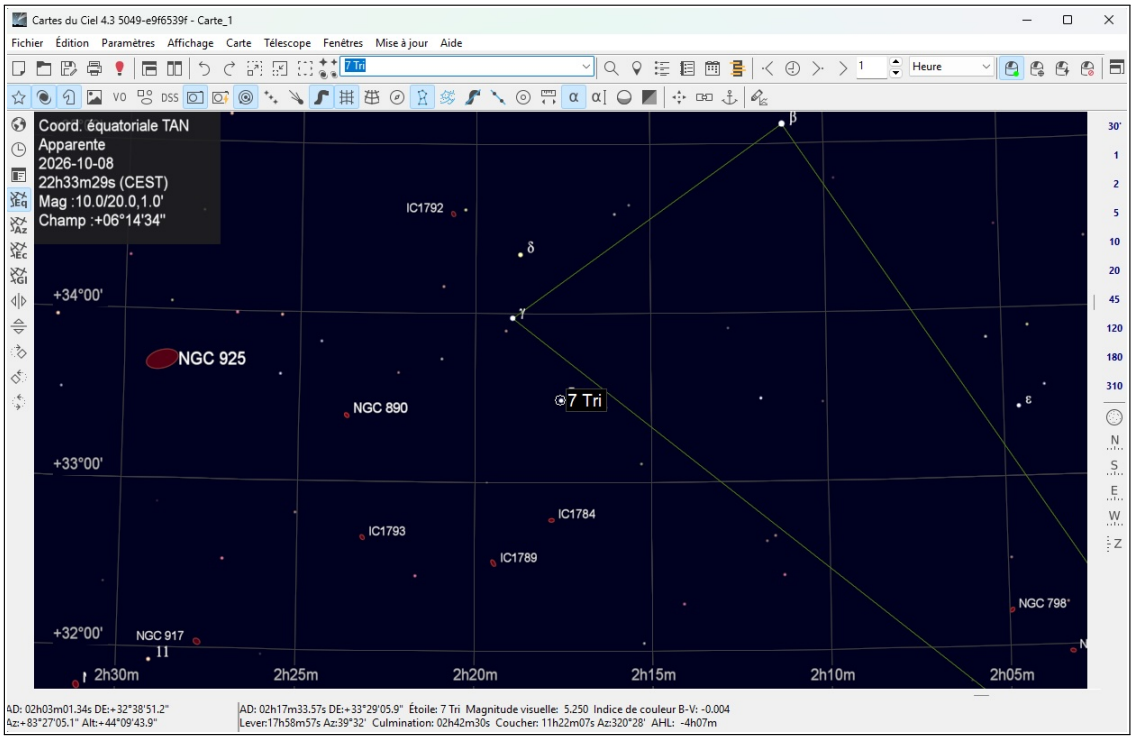



**Figure 4.** Selecting the object to point at in *Carte du Ciel*.

Then simply click the 'Goto' button in *CCDciel* (figure 5).

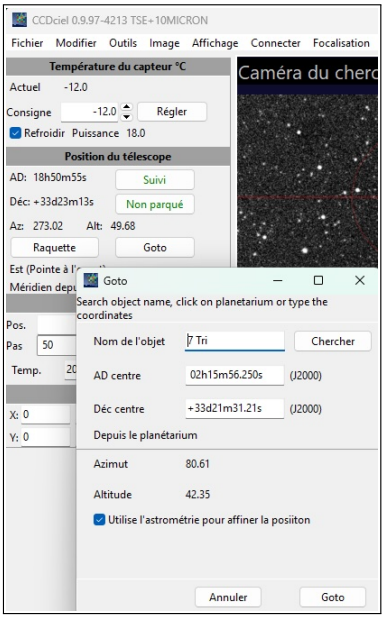



**Figure 5.** The automatic pointing function of *CCDciel*.

The star 7 Tri was not chosen at random: it is fairly bright and, above all, of spectral type A, with strong Balmer lines, ideal for a first acquisition. Vega is of the same spectral type, but it is generally too bright for a good-quality low-resolution acquisition, owing to the risk of saturation.

## 4 Locating the slit

Begin by taking an image of the slit with the `TinyStar'Ex` guiding camera. To do so, the sky must illuminate the slit more or less uniformly. There are three options:

- a short exposure at twilight, the best option;
- an exposure of a few tens of seconds in the middle of the night, under a light-polluted sky;
- failing that, illuminating the telescope aperture as uniformly as possible, the least satisfactory option.

The slit then appears as a dark line on a bright background. It should be sharp, just as you left it after the bench adjustments.

Nevertheless, check this sharpness, zooming well into the image if necessary: be demanding (figure 6). Note that we are guiding in declination here, which is unusual. If you guide in right ascension, the slit may appear vertical. Its orientation depends on how the slit is mounted in the guiding system, on the orientation of the spectrograph on the tube, and on that of the tube itself: choose the configuration that suits you best.

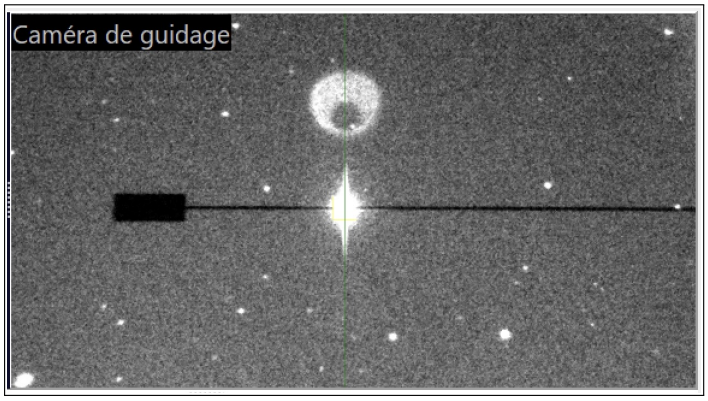



**Figure 6.** Exposure set so as to make the image of the slit visible, even at the cost of temporarily saturating that of the target. A reticle marks the position where objects must be placed from now on.

Next, note the position of the slit to the nearest pixel, on both the horizontal and the vertical axis. These coordinates are valuable: it is on this precise point of the image that the star must be brought, then held there throughout the exposure. This is where software such as *Prism* or *CCDciel* shows its full power: it stores these coordinates and displays them on screen as a reticle.

## 5 Focusing

The images are unlikely to be sharp from the very first exposure. The essential step now is to focus the stars accurately on the slit, using the telescope's focuser. A sharp slit and sharp stars: that is the recipe for success.

In the example of figure 7, the star is clearly out of focus. Two symptoms give this away: on the guiding camera, the stars look abnormally spread out; on the acquisition camera, the spectrum trace is also wide, with a dark band running along its middle, which is the shadow of the secondary mirror.

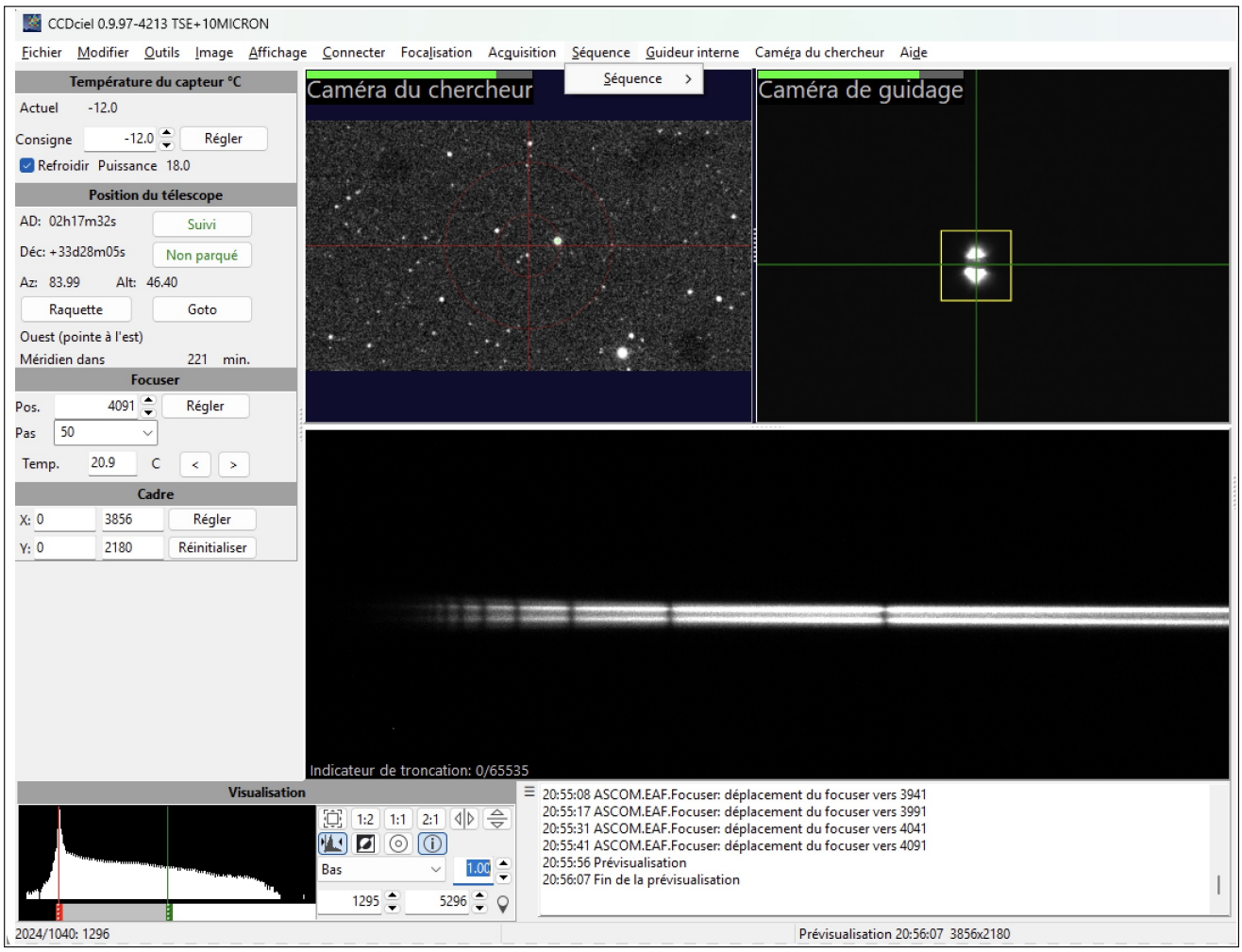



**Figure 7.** Guiding image and spectrum of a poorly focused star. Do not be caught out: to judge the result, saturate neither image.

Adjust the telescope's focus to obtain the narrowest possible spectrum (figure 8). The stars in the guiding field should then be in focus as well[1].

[1] This is one of the great strengths of `TinyStar'Ex`: a single focus adjustment, that of the telescope, brings both the spectral channel and the guiding channel into focus, with no additional adjustment and no hesitation. Moreover, the TSE optics are of such quality that the spectrum trace remains perfectly focused from the ultraviolet to the deep red. In use, everything becomes simple.

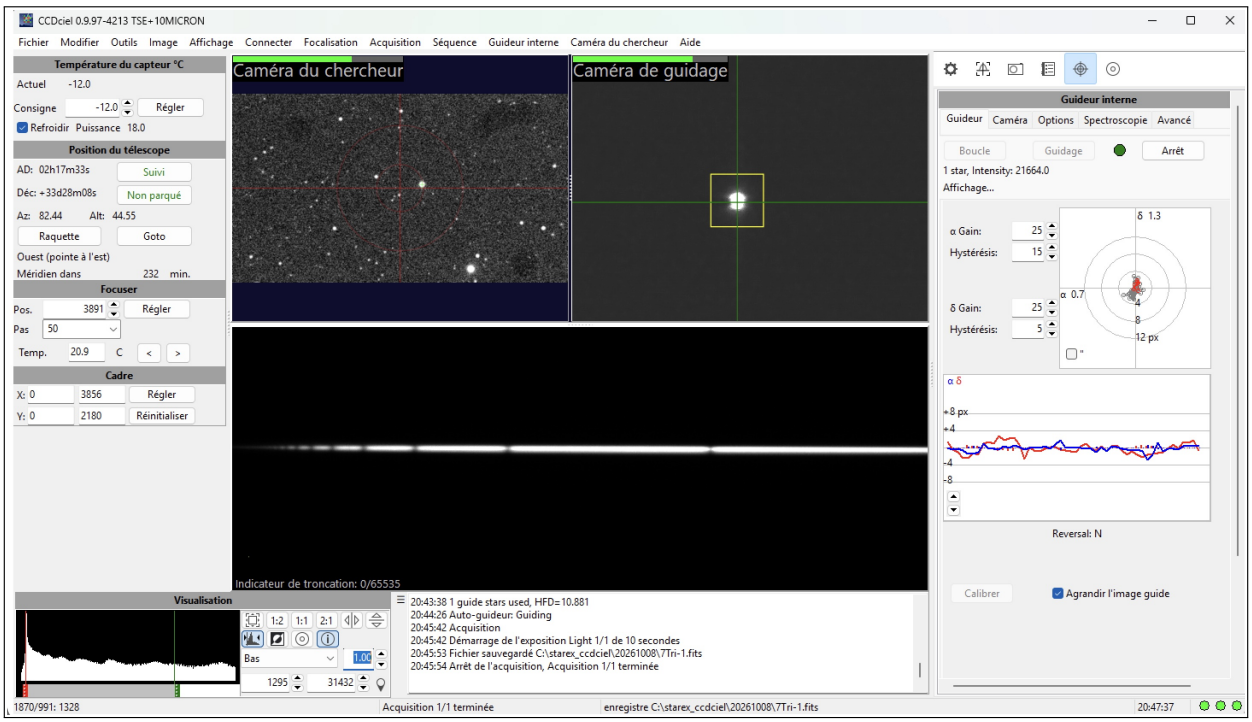



**Figure 8.** Appearance of the spectrum once the star images are focused on the entrance slit.

During this focusing operation, it is important that the spectrum trace recorded by the detector is not saturated. Most acquisition software includes a saturation indicator: remember to turn it on.

Note, on the right of figure 8, that autoguiding has been started. *CCDciel* offers many tools for centring and guiding the target, on or off the slit. This is one of the great strengths of this software: once you have tried it, there is no going back!

Normally, the lines of your TSE spectrograph are sharp; the broadening seen in the case of 7 Tri is natural: it is intrinsic to the star. Several techniques allow you to check this. One of them is to acquire the spectrum of a small neon nightlight waved in front of the telescope aperture (figure 9): if its monochromatic lines appear sharp, with well-defined edges, all is well. If not, refocus the spectrograph on the telescope itself: open the access hatch to the camera-lens tube, then rotate this tube one way or the other, in steps of 10° at first, then progressively finer.



**Figure 9.** A neon nightlight is waved in front of the telescope during an exposure of 30 to 60 seconds. This gives an emission-line spectrum on which the quality of focus can easily be judged with precision.

## 6 Routine acquisition

Your instrument is now operational (figure 10). Enjoy using it!



**Figure 10.** The 250 mm f/4 telescope, fitted with TinyStar'Ex, in operation (near the town of Antibes, hence the light pollution).

Spectrum processing is not covered here; it remains simple with the right tools, a little method and sound advice. Download the specINTI software from: https://solex.astrosurf.com/specinti1_fr.html. Then consult the quick-start guide, which is based precisely on TinyStar'Ex spectra:

https://buil.astrosurf.com/specinti4/specINTI_demarrage_rapide.pdf